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 Studierichting: Informatica
 Oefeningenreeks 3H nr. 8.3

$$r = \sin 3\theta$$

$$\text{dom } f = \mathbb{R}$$

$$\text{Periode } T = \frac{2\pi}{3}$$

We beperken ons tot het interval $\left[0, \frac{2\pi}{3}\right]$

Nulpunten:

$$3\theta = n\pi \Leftrightarrow \theta = \frac{n}{3}\pi$$

In het interval zijn de nulpunten: $\left\{0, \frac{\pi}{3}, \frac{2\pi}{3}\right\}$

Afgeleide:

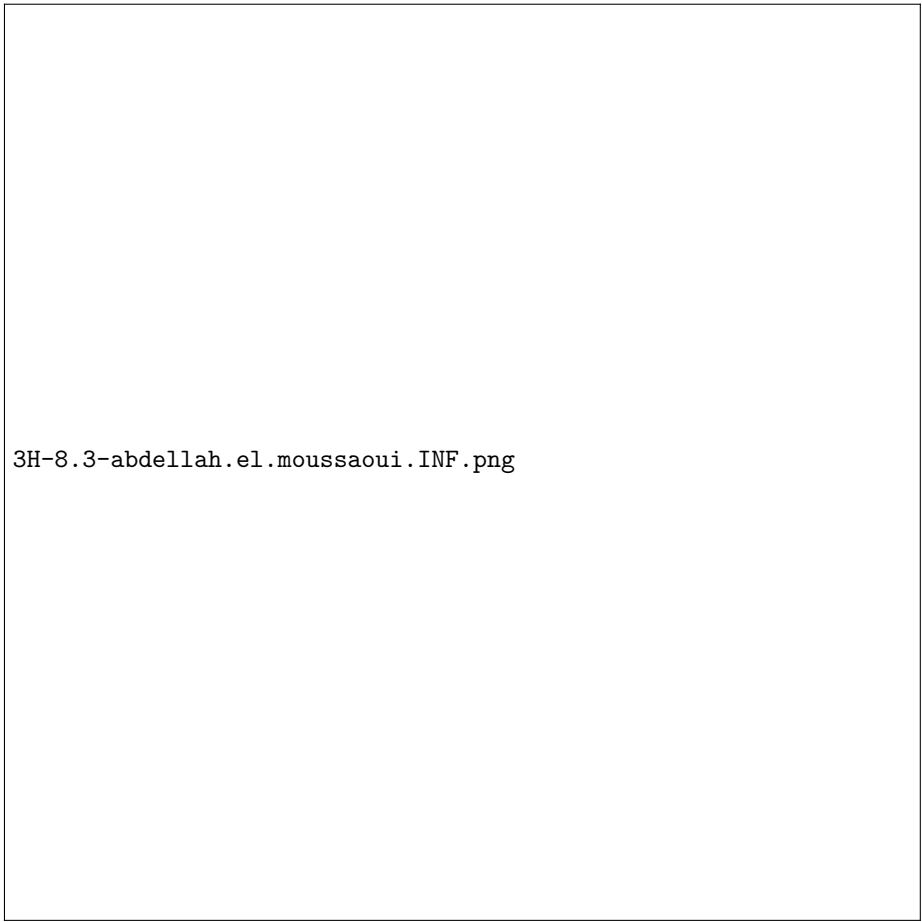
$$r' = 3 \cos 3\theta$$

$$r' = 0 \Leftrightarrow \theta = \frac{\pi}{6} + n \cdot \frac{\pi}{3} \Leftrightarrow \theta \in \left\{\frac{\pi}{6}, \frac{\pi}{2}\right\}$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$
r	0	+	+	0	-
r'	+	+	0	-	-
r	$\nearrow$	$\nearrow$	M	$\searrow$	$\searrow$
			1		
				m	$\nearrow$
				-1	$\nearrow$

Grafiek:

References



3H-8.3-abdellah.el.moussaoui.INF.png